

DILAVAR AFREED.E

MACHINE LEARNING ENGINEER

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ABOUT ME

Dynamic and results-driven Machine Learning Engineer with expertise in Computer Vision, Natural Language Processing (NLP), Large Language Models (LLMs) encompassing training, fine-tuning, quantization, and deployment, as well as Generative AI. Proficient in delivering innovative solutions to complex problems through cutting-edge technologies.

WORK EXPERIENCE

MACHINE LEARNING ENGINEER | Feathersoft Info Solutions, Kochi

Aug 2023 – Present

- Developed and fine-tuned computer vision models for tasks such as image segmentation, object detection, and tracking, significantly improving model accuracy and performance.
- Implemented advanced machine learning techniques and optimized models using TensorFlow and PyTorch, enhancing real-time processing capabilities and reducing latency
- Leveraged large language models for generating contextually relevant responses.
- Created a natural language generation model to power a conversational chatbot for customer service.

JR. DATA SCIENTIST | Techolas Technologies, Calicut

Sept 2021 – Aug 2022

- Automated data collection processes from various sources and conducted pre-processing of structured and unstructured data to ensure data quality.
 - Built predictive models and machine learning algorithms to solve business challenges and generate insights from data.
 - Presented findings through data visualization techniques, enabling informed decision-making across teams.
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TECHNICAL SKILLS

Natural Language Processing (NLP)

- Transformers (BERT, GPT, etc.)
- Large Language Models (LLMs) – Fine-tuning & Quantization
- Generative AI (Text Generation, Summarization, Chatbots)
- LangChain & Retrieval-Augmented Generation (RAG)
- Prompt Engineering & Context Optimization

Computer vision

- OpenCV & Vision Transformers
- Object Detection & Tracking (YOLO, Faster R-CNN)
- Generative Adversarial Networks (GANs)
- Image Classification & Segmentation

Machine Learning & Deep Learning

- Python, SQL
- Supervised & Unsupervised Learning
- Transfer Learning & Model Optimization
- TensorFlow, PyTorch, Scikit-learn
- Model Deployment & MLOps (Docker, APIs, Hugging Face Hub)
- Data Preprocessing & Feature Engineering
- Version Control (Git/GitHub)
- Containerization (Docker)

PROJECTS

Medical Report Generation with GraphRAG

- Built an AI-driven medical report generation system integrating graph databases, vector databases, and intelligent agents for precise knowledge retrieval and synthesis.
- Designed a graph-based medical knowledge representation to map relationships between symptoms, diagnoses, and treatments for context-aware reasoning.
- Developed micro-agents for patient data extraction and processing, orchestrated by a manager agent for efficient insight integration.
- Implemented vector search for semantic retrieval and LLM-based generation for clinical narratives.
- Optimized pipeline via iterative testing and collaborated with healthcare professionals to ensure accuracy and clinical usability.
- Skills Used - Knowledge Graph, Vector Databases, LLMs, RAG, LangChain, Context Engineering

Interactive Avatar with Lip-Synced Chatbot Responses

- Built an interactive avatar system delivering real-time, lip-synced video responses for chatbot interactions.
- Integrated f5-TTS for converting chatbot responses into natural audio and Dinet model for precise lip-syncing.
- Combined OpenCV with generative models for responsive avatar rendering and seamless communication.
- Deployed system via FastAPI for real-time performance and scalable chatbot integration.
- Skills Used - Text-to-Speech (f5-TTS), Lip Sync (Dinet), OpenCV, LLM Integration, FastAPI

Custom Object Tracking and Counting System

- Developed a robust object tracking and counting pipeline to eliminate double-counting when objects re-entered the frame.
- Designed a custom algorithm combining Kalman Filter and IoU for accurate multi-object tracking.
- Fine-tuned YOLOv8 for high-precision object detection and integrated with OpenCV for real-time performance.
- Implemented re-entry handling logic to ensure accurate counting across multiple frame passes.
- Skills Used - YOLOv8, Kalman Filter, IoU, Object Tracking, OpenCV

Object Inpainting using GAN

- Built a GAN-based object inpainting model for seamless object removal and background reconstruction.
- Utilized SAM (Segment Anything Model) for precise object segmentation and mask generation.
- Developed an interactive interface for intuitive object selection and editing.
- Enhanced image realism using advanced loss functions to ensure consistent textures and color blending.
- Skills Used - GANs, SAM, Image Segmentation, OpenCV, TensorFlow, Computer Vision

EDUCATION

B.Tech Textile Technology

2016-2020

PCET Coimbatore

Tamilnadu

RESEARCH DOMAINS

- Building large language models from scratch.
- Finetuning Large language models.
- Quantizing LLM models for inference on low end devices.
- Attention in LLMs
- Long context LLM modelling.